

C# COMPLETE GUIDE: CHAR, STRING, STRINGBUILDER

CHAR FUNCTIONS

IsLetter()
IsDigit()
IsLetterOrDigit()
IsWhiteSpace()
IsUpper()
IsLower()
IsNumber()
IsPunctuation()
IsSymbol()
IsControl()
ToUpper()
ToLower()
GetNumericValue()
Parse()
TryParse()
Compare()
GetUnicodeCategory()

Example:

```
char ch = 'A';  
Console.WriteLine(char.IsLetter(ch));
```

STRING FUNCTIONS

Length
Trim()
TrimStart()
TrimEnd()
ToUpper()
ToLower()
Contains()
StartsWith()
EndsWith()
IndexOf()
LastIndexOf()
Substring()
Replace()
Split()
Join()
Equals()
Compare()
IsNullOrEmpty()
IsNullOrWhiteSpace()

Example:

```
string text = "SoftwareEngineering";  
Console.WriteLine(text.Substring(0, 8));
```

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STRINGBUILDER FUNCTIONS

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Constructors:

StringBuilder()
StringBuilder(string)
StringBuilder(int capacity)
StringBuilder(string, int capacity)

Properties:

Length
Capacity
MaxCapacity

Methods:

Append()
AppendLine()
AppendFormat()
AppendJoin()
Insert()
Remove()
Replace()
Clear()
ToString()
ToString(int, int)
EnsureCapacity()
CopyTo()
Equals()
GetHashCode()

Indexer:

sb[index]

Example:

```
StringBuilder sb = new StringBuilder("Hello");  
sb.Append(" World");  
Console.WriteLine(sb.ToString());
```

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IMPORTANT THEORY POINTS

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- string is immutable
- StringBuilder is mutable
- char is a struct (2 bytes UTF-16)
- StringBuilder improves performance in loops
- StringBuilder is not thread-safe
- string is thread-safe (immutable)